Time Series Analysis & Classification

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Chapter IX: Seasonal ARIMA Models

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2. Seasonal Moving Average (MA) Models
3. Seasonal Autoregressive (AR) Models
4. Seasonal $ARMA$ Models
5. Multiplicative Seasonal $ARMA$ Models
6. Nonstationary seasonal $ARIMA$ ($SARIMA$) models
7. Modelling Example

Introduction

Introduction to Seasonal Models

- Time series often exhibit seasonality (repeating patterns).
- Examples: Monthly sales, daily temperatures.
- Need specialized models to capture these patterns.

Discussion: Seasonality can significantly impact forecasting accuracy. Understanding seasonal patterns is crucial for effective time series analysis.

Seasonal Moving Average (MA) Models

Seasonal $MA(Q)_s$ Model

- Model:

$$Y_t = e_t - \Theta_1 e_{t-s} - \Theta_2 e_{t-2s} - \cdots - \Theta_Q e_{t-Qs}$$

  - Y_t: Time series value at time t
  - e_t: White noise error term
  - Θ_i: Seasonal MA parameters
  - s: Seasonal period

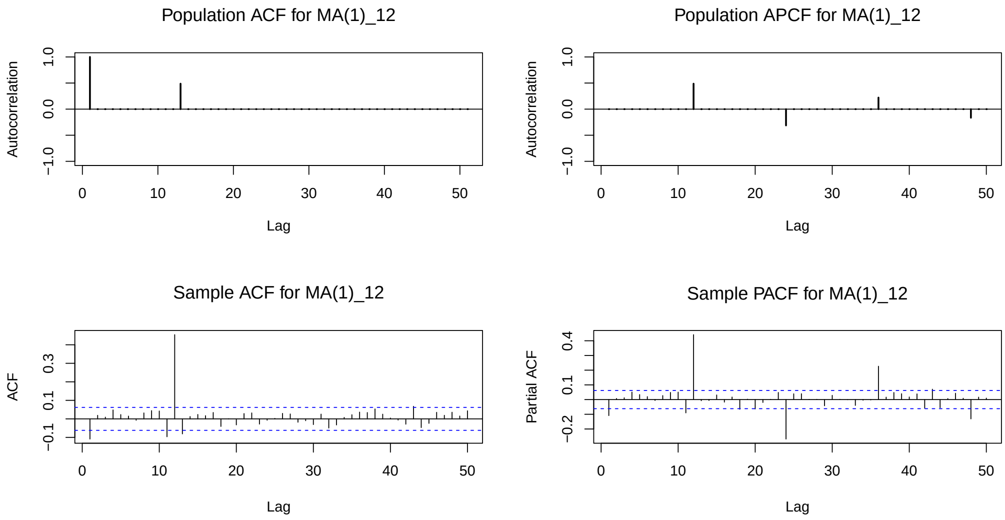
Comment: This model captures seasonal effects by incorporating past error terms at seasonal lags.

Example: $MA(1)_{12}$

- Model: $Y_t = e_t - \Theta e_{t-12}$
- ACF: Nonzero only at lag $k = 12$
- PACF: Decays at seasonal lags $k = 12, 24, 36, \ldots$

Discussion: The $MA(1)_{12}$ model is useful for series with a single seasonal spike in the ACF.

Example: $MA(1)_{12}$



Example: $MA(2)_{12}$

- Model: $Y_t = e_t - \Theta_1 e_{t-12} - \Theta_2 e_{t-24}$
- Autocovariance:

$$\gamma_k = \begin{cases} \sigma_e^2(1+\Theta_1^2+\Theta_2^2), & k=0 \\ (-\Theta_1+\Theta_1\Theta_2)\sigma_e^2, & k=12 \\ -\Theta_2\sigma_e^2, & k=24 \\ 0, & \text{otherwise} \end{cases}$$

- ACF:

$$\rho_k = \frac{\gamma_k}{\gamma_0} = \begin{cases} 1, & k=0 \\ \frac{-\Theta_1+\Theta_1\Theta_2}{1+\Theta_1^2+\Theta_2^2}, & k=12 \\ \frac{-\Theta_2}{1+\Theta_1^2+\Theta_2^2}, & k=24 \\ 0, & \text{otherwise} \end{cases}$$

- ACF has the same form as nonseasonal $MA(2)$

Example: $MA(2)_{12}$

Comment: The $MA(2)_{12}$ model extends the seasonal effect to two lags, capturing more complex seasonal patterns. Note that the ACF has nonzero autocorrelations at seasonal lags $k = 12$ and $k = 24$, similar to nonseasonal $MA(2)$.

$MA(Q)_s$: Backshift Notation

- $Y_t = (1 - \Theta_1 B^s - \Theta_2 B^{2s} - \cdots - \Theta_Q B^{Qs}) e_t = \Theta_Q(B^s) e_t$
- B: Backshift operator ($B^s e_t = e_{t-s}$)
- $\Theta_Q(B^s)$: Seasonal MA characteristic operator

$MA(Q)_s$: Stationarity and Invertibility

- Stationarity: Always
- Invertibility: Roots of $\Theta_Q(B^s)$ exceed 1 in absolute value.

Seasonal Autoregressive (AR) Models

Seasonal $AR(p)_s$ Model

- Model:
$$Y_t = \Phi_1 Y_{t-s} + \Phi_2 Y_{t-2s} + \cdots + \Phi_P Y_{t-Ps} + e_t$$
- Y_t: Time series value at time t
- e_t: White noise error term
- Φ_i: Seasonal AR parameters
- s: Seasonal period

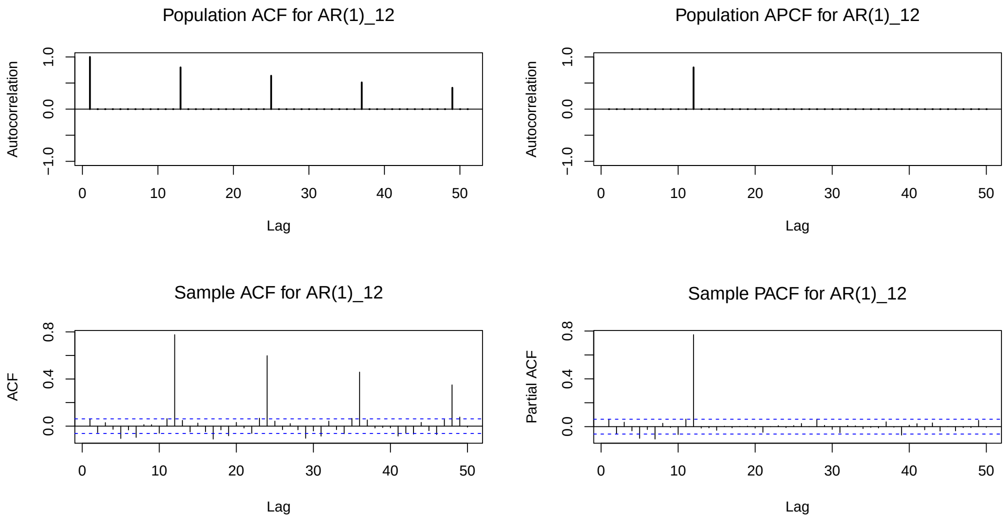
Discussion: AR models capture autoregressive effects at seasonal lags, useful for series with strong seasonal dependencies.

Example: $AR(1)_{12}$

- Model: $Y_t = \Phi\, Y_{t-12} + e_t$
- Stationary: $-1 < \Phi < 1$
- ACF: $\rho_k = \begin{cases} \Phi^{k/12}, & k = 0, 12, 24, 36, \ldots \\ 0, & \text{otherwise} \end{cases}$
- PACF: Nonzero only at lag 12

Comment: The $AR(1)_{12}$ model is suitable for series with a single seasonal spike in the PACF. The ACF is 0 at lags that are not multiples of $s = 12$.

Example: $AR(1)_{12}$



Example: $AR(2)_{12}$

- Model: $Y_t = \Phi_1 Y_{t-12} + \Phi_2 Y_{t-24} + e_t$
- ACF: Exponential decay or damped sinusoidal
- PACF: Nonzero at lags 12 and 24

Discussion: The $AR(2)_{12}$ model captures more complex seasonal autoregressive patterns. It behaves like a nonseasonal $AR(2)$ at the seasonal lags.

$AR(P)_s$: Backshift Notation

- $(1 - \Phi_1 B^s - \Phi_2 B^{2s} - \cdots - \Phi_P B^{Ps})\, Y_t = e_t \implies \Phi_P(B^s)\, Y_t = e_t$
- $\Phi_P(B^s)$: Seasonal AR characteristic operator

$AR(P)_s$: Stationarity and Invertibility

- Stationarity: Roots of $\Phi_P(B^s)$ exceed 1 in absolute value
- Invertibility: Always

Seasonal $ARMA$ Models

Seasonal $ARMA(P, Q)_s$ Model

- Model:

$$Y_t = \Phi_1 Y_{t-s} + \cdots + \Phi_P Y_{t-Ps} + e_t - \Theta_1 e_{t-s} - \cdots - \Theta_Q e_{t-Qs}$$

- Using Backshift notation: $\Phi_P(B^s) Y_t = \Theta_Q(B^s) e_t$

Comment: Combines AR and MA components to capture both autoregressive and moving average effects at seasonal lags. This is a seasonal analogue of the nonseasonal $ARMA(p, q)$ process.

Seasonal $ARMA(P, Q)_s$: Stationarity Invertibility

- Stationarity: Roots of $\Phi_P(B^s)$ exceed 1 in absolute value (or modulus)
- Invertibility: Roots of $\Theta_Q(B^s)$ exceed 1 in absolute value (or modulus)

ACF and PACF Summary

Process	ACF	PACF
$AR(P)_s$	Tails off at lags ks	Cuts off after lag Ps Non-zero at lags ks
$MA(Q)_s$	Cuts off after lag Qs Non-zero at lags ks	Tails off at lags ks
$ARMA(P, Q)_s$	Tails off at lags ks	Tails off at lags ks

Connections to Nonseasonal Models

- A seasonal $MA(2)_{12}$ is mathematically equivalent to a nonseasonal $MA(24)$ with specific parameter restrictions.
- A seasonal $AR(1)_{12}$ is mathematically equivalent to a nonseasonal $AR(12)$ with specific parameter restrictions.
- An $ARMA(P, Q)_s$ is mathematically equivalent to a nonseasonal $ARMA(Ps, Qs)$ with parameter restrictions.

Implication: We can leverage existing methods for specifying, fitting, diagnosing, and forecasting seasonal models.

Limitations and Extensions

- $ARMA(P, Q)_s$ models alone are limited in application for stationary processes.

Discussion: These models incorporate autocorrelation at seasonal lags and nowhere else.

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Extension: Combine seasonal $ARMA(P, Q)_s$ models with traditional nonseasonal $ARMA(p, q)$ models to create a larger class of models applicable for use with stationary processes that exhibit seasonality (Multiplicative Seasonal $ARMA$).

Multiplicative Seasonal $ARMA$ Models

Introduction

- Time series models often exhibit both seasonal and nonseasonal dependencies.
- Multiplicative models allow for an interaction between these components.

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- Multiplicative models allow for an interaction between these components.
- This section focuses on $MA(1) \times MA(1)_{12}$ and $MA(1) \times AR(1)_{12}$ processes.

Multiplicative Seasonal $ARMA$ Model

Definition: A multiplicative seasonal $ARMA$ model combines a nonseasonal $ARMA(p, q)$ process with a seasonal $ARMA(P, Q)_s$ process:

$$\phi(B)\Phi_P(B^s)\,Y_t = \theta(B)\Theta_Q(B^s)\,e_t$$

where:

- $\phi(B)$ and $\theta(B)$ are the nonseasonal AR and MA polynomials, respectively.
- $\Phi_P(B^s)$ and $\Theta_Q(B^s)$ are the seasonal AR and MA polynomials, respectively, with seasonal period s.

Key Properties

- Flexibility: This model family is very flexible for modeling stationary seasonal processes.

- Equivalence: An $ARMA(p, q) \times ARMA(P, Q)_s$ process is equivalent to a nonseasonal $ARMA$ process with modified AR and MA characteristic operators.

$$\phi^*(B) = \phi(B)\Phi_P(B^s), \qquad \theta^*(B) = \theta(B)\Theta_Q(B^s)$$

- Stationarity and Invertibility: Conditions depend on the roots of $\phi^*(B)$ and $\theta^*(B)$.

- Established Methods: We can use already-established methods to specify, fit, diagnose, and forecast seasonal stationary models.

MA(1) $\times$ MA(1)$_{12}$ Model

- Suppose $\{e_t\}$ is a zero mean white noise process with variance σ_e^2.
- Nonseasonal $MA(1)$:

$$Y_t = e_t - \theta e_{t-1} = (1 - \theta B) e_t$$

- Seasonal $MA(1)_{12}$:

$$Y_t = e_t - \Theta e_{t-12} = (1 - \Theta B^{12}) e_t$$

- Multiplying the two:

$$Y_t = (1 - \theta B)(1 - \Theta B^{12}) e_t$$

Autocorrelation Properties of MA(1) $\times$ MA(1)$_{12}$

- The nonseasonal $MA(1)$ model has nonzero autocorrelation only at lag $k = 1$.
- The seasonal $MA(1)_{12}$ model has nonzero autocorrelation only at lag $k = 12$.
- The multiplicative combination introduces additional nonzero autocorrelation at lags $k = 11$ and $k = 13$.
- The process is equivalent to an $MA(13)$ model with:

$$\theta_1 = \theta, \quad \theta_{12} = \Theta, \quad \theta_{13} = -\theta\Theta,$$
$$\theta_2 = \theta_3 = \cdots = \theta_{11} = 0.$$

MA(1) $\times$ AR(1)$_{12}$ Model

- Combining:

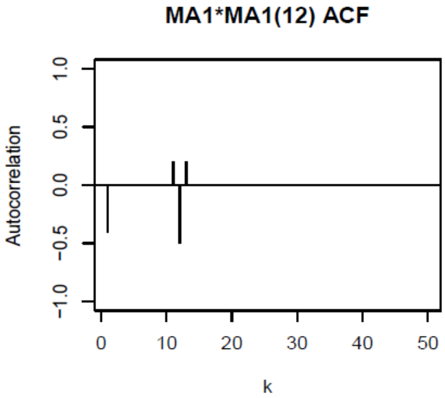
$$Y_t = e_t - \theta e_{t-1}, \quad \text{and} \quad Y_t = \Phi Y_{t-12} + e_t$$

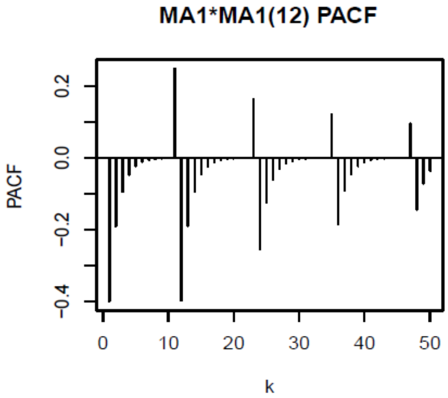
- Results in:

$$Y_t = \Phi Y_{t-12} + e_t - \theta e_{t-1}$$

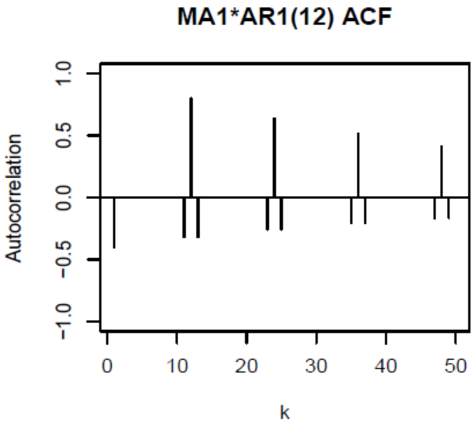
- This process exhibits seasonal AR structure with additional MA-type effects at $k = 1$, $k = 11$, and $k = 13$.

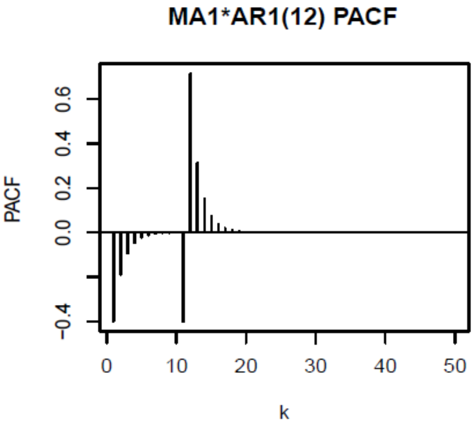
MA(1) $\times$ MA(1)$_{12}$





MA(1) $\times$AR(1)$_{12}$





Discussion and Insights

- Multiplicative models capture interactions between seasonal and nonseasonal effects.

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- Multiplicative models capture interactions between seasonal and nonseasonal effects.
- Stationarity and invertibility conditions must be checked for valid model specification.
- Model selection should be based on diagnostic tests, autocorrelation functions, and practical interpretability.
- These models provide a foundation for forecasting in seasonal time series data.

Summary

The multiplicative seasonal (stationary) $ARMA(p,q) \times ARMA(P,Q)_s$ family of models:

$$\phi(B)\Phi_P(B^s)\,Y_t = \theta(B)\Theta_Q(B^s)\,e_t.$$

is a flexible class of time series models for stationary seasonal processes.

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is a flexible class of time series models for stationary seasonal processes. The next step is to extend this class of models to handle two types of nonstationarity:

- Nonseasonal nonstationarity over time (e.g., increasing linear trends, etc.)
- Seasonal nonstationarity, that is, additional changes in the seasonal mean level, even after possibly adjusting for nonseasonal stationarity over time.

Nonstationary seasonal $ARIMA$ ($SARIMA$) models

Example: Nonseasonal Differencing

- Suppose that we have a stochastic process defined by:

$$Y_t = S_t + e_t, \quad S_t = S_{t-12} + u_t.$$

That is, $\{S_t\}$ is a zero mean random walk with period $s = 12$.

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- Taking first differences:

$$\nabla Y_t = S_{t-12} - S_{t-13} + u_t - u_{t-1} + e_t - e_{t-1}.$$

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- Taking first differences:

$$\nabla Y_t = S_{t-12} - S_{t-13} + u_t - u_{t-1} + e_t - e_{t-1}.$$

Issue: ∇Y_t is still nonstationary across seasons, i.e, across time points $t = 12k$.

Seasonal Differencing

- Definition: Seasonal difference operator:

$$\nabla_s Y_t = Y_t - Y_{t-s} = (1 - B^s)\, Y_t.$$

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- Example for monthly data ($s = 12$):

$$\nabla_{12} Y_t = Y_t - Y_{t-12}.$$

Example: Seasonal Differencing

- Applying seasonal differencing:

$$\nabla_{12} Y_t = S_t - S_{t-12} + e_t - e_{t-12} = u_t + e_t - e_{t-12}.$$

- Result:

Example: Seasonal Differencing

- Applying seasonal differencing:

$$\nabla_{12} Y_t = S_t - S_{t-12} + e_t - e_{t-12} = u_t + e_t - e_{t-12}.$$

- Result: A stationary seasonal $MA(1)_{12}$ process.

Combining Differencing

Two types of differencing: We can combine the two types of differencing to deal with complexe forms of nonstationarity

$$\nabla \nabla_{12} Y_t =$$

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$$\nabla\nabla_{12} Y_t = (1 - B)(1 - B^{12}) Y_t.$$

Expanding:

$$\nabla\nabla_{12} Y_t = Y_t - Y_{t-1} - Y_{t-12} + Y_{t-13}.$$

Example

The model described previously could also be generalized to account for a nonseasonal, slowly changing stochastic trend:

$$
\begin{aligned}
Y_t &= M_t + S_t + e_t \\
S_t &= S_{t-s} + \varepsilon_t \\
M_t &= M_{t-1} + \xi_t
\end{aligned}
$$

where $\{e_t\}$, $\{\varepsilon_t\}$, and $\{\xi_t\}$ are mutually independent white noise series.

Example

- Here we take both a seasonal difference and an ordinary nonseasonal difference to obtain:

$$\begin{aligned}\nabla\nabla_s Y_t &= \nabla(M_t - M_{t-s} + \varepsilon_t + e_t - e_{t-s}) \\ &= (\xi_t + \varepsilon_t + e_t) - (\varepsilon_{t-1} + e_{t-1}) - e_{t-s-1}\end{aligned}$$

- The process defined here is stationary and has nonzero autocorrelation only at lags 1, $s-1$, s, and $s+1$, which agrees with the autocorrelation structure of the multiplicative seasonal model $ARMA(0,1) \times (0,1)$ with seasonal period s.

Example

- Here we take both a seasonal difference and an ordinary nonseasonal difference to obtain:

$$\begin{aligned}\nabla\nabla_s Y_t &= \nabla(M_t - M_{t-s} + \varepsilon_t + e_t - e_{t-s})\\ &= (\xi_t + \varepsilon_t + e_t) - (\varepsilon_{t-1} + e_{t-1}) - e_{t-s-1}\end{aligned}$$

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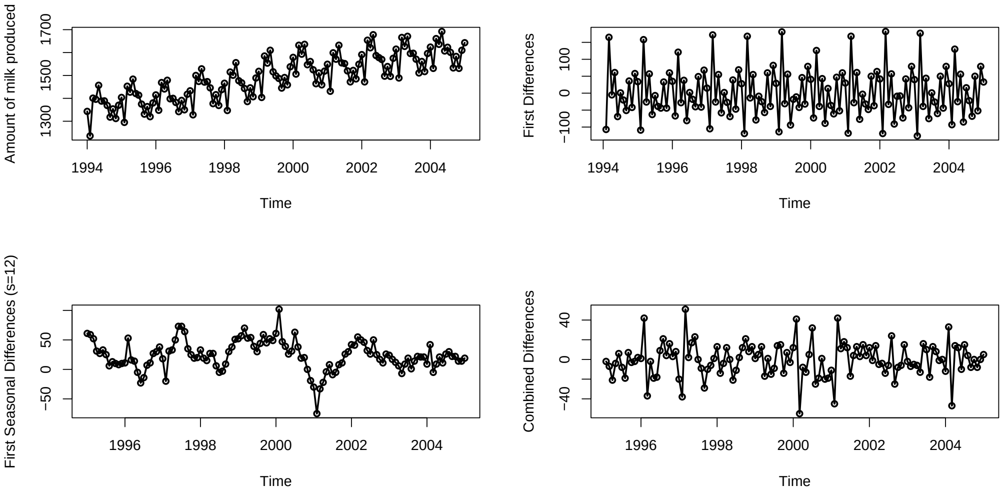
- The process defined here is stationary and has nonzero autocorrelation only at lags 1, $s-1$, s, and $s+1$, which agrees with the autocorrelation structure of the multiplicative seasonal model $ARMA(0,1) \times (0,1)$ with seasonal period s.

Example



Amount of milk produced
1300
1500
1700
1994
1996
1998
2000
2002
2004
Time
First Differences
−100
0
100
1994
1996
1998
2000
2002
2004
Time
First Seasonal Differences (s=12)
−50
0
50
1996
1998
2000
2002
2004
Time
Combined Differences
−40
0
40
1996
1998
2000
2002
2004
Time

Definition of $SARIMA$ Models

Model:

$$\phi(B)\Phi_P(B^s)\nabla^d\nabla_s^D Y_t = \theta(B)\Theta_Q(B^s)e_t.$$

Where:

- d: Nonseasonal differencing order.
- D: Seasonal differencing order.
- $\phi(B)$, $\theta(B)$: Nonseasonal AR and MA polynomials.
- $\Phi_P(B^s)$, $\Theta_Q(B^s)$: Seasonal AR and MA polynomials.

Definition of $SARIMA$ Models

- Nonseasonal AR and MA operators:

$$\phi(B) = (1 - \phi_1 B - \phi_2 B^2 - \cdots - \phi_p B^p)$$

$$\theta(B) = (1 - \theta_1 B - \theta_2 B^2 - \cdots - \theta_q B^q)$$

- Seasonal AR and MA operators:

$$\Phi_P(B^s) = 1 - \Phi_1 B^s - \Phi_2 B^{2s} - \cdots - \Phi_P B^{Ps}$$

$$\Theta_Q(B^s) = 1 - \Theta_1 B^s - \Theta_2 B^{2s} - \cdots - \Theta_Q B^{Qs}$$

- Differencing operator:

$$\nabla^d \nabla_s^D Y_t = (1 - B)^d (1 - B^s)^D Y_t.$$

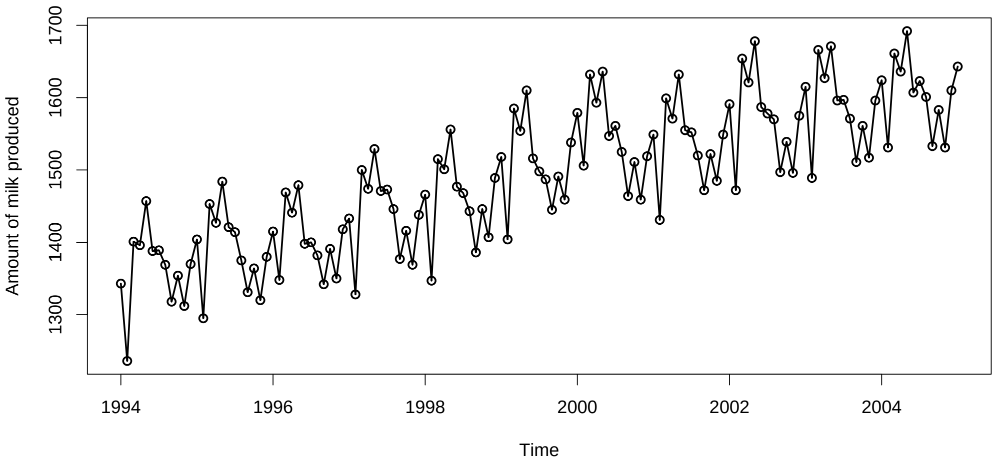
Model Interpretation

- d denotes the number of nonseasonal differences. Typically, $d = 1$ or $d = 2$ ensures nonstationarity removal.
- D denotes the number of seasonal differences. Seasonal differencing $D = 1$ is often sufficient.
- $SARIMA$ models are effective for capturing both trend and seasonality.

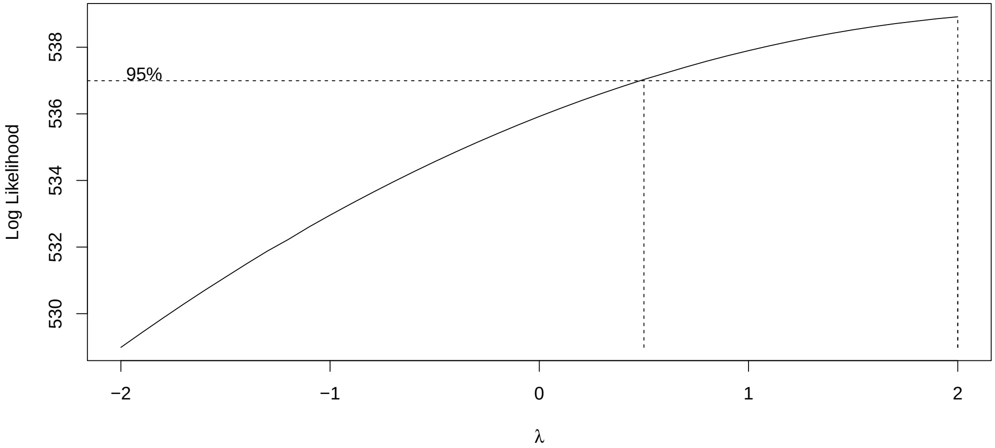
Modelling Example

- We consider Milk production data (package `TSA`), which represents the monthly U.S. milk production (in millions of pounds) from January, 1994 to December, 2005.

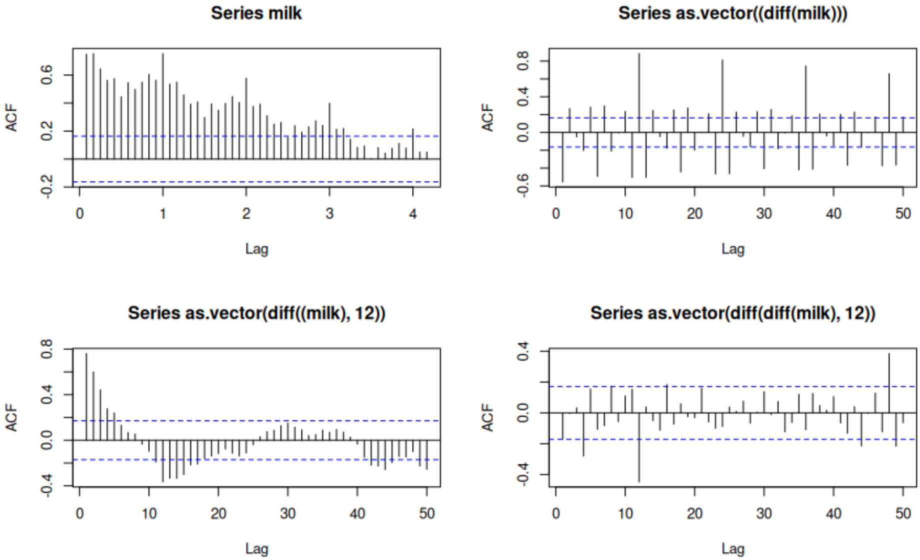
Step 1: Data Plot



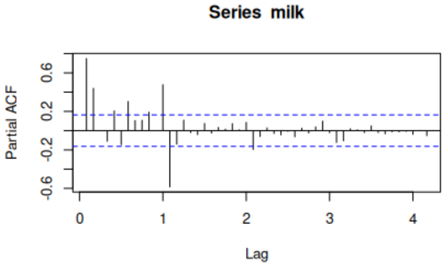
Step 2: Pre-Processing of Data

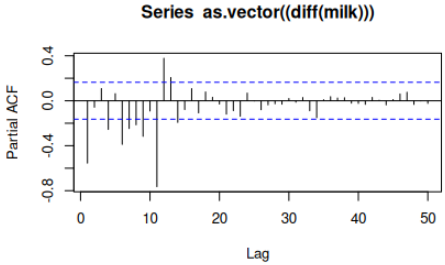


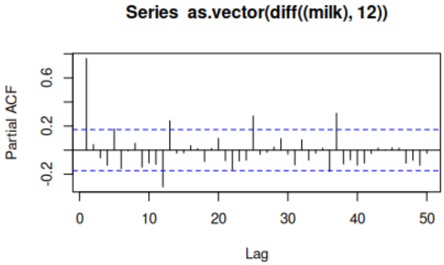
ACF Plots

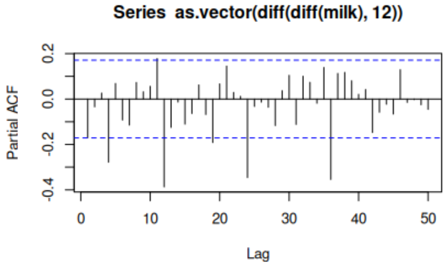

Series milk
ACF
0.6
0.2
-0.2
0
1
2
3
4
Lag
Series as.vector((diff(milk)))
ACF
0.8
0.4
0.0
-0.6
0
10
20
30
40
50
Lag
Series as.vector(diff((milk), 12))
ACF
0.8
0.4
0.0
-0.4
0
10
20
30
40
50
Lag
Series as.vector(diff(diff(milk), 12))
ACF
0.4
0.0
-0.4
0
10
20
30
40
50
Lag

PACF Plots

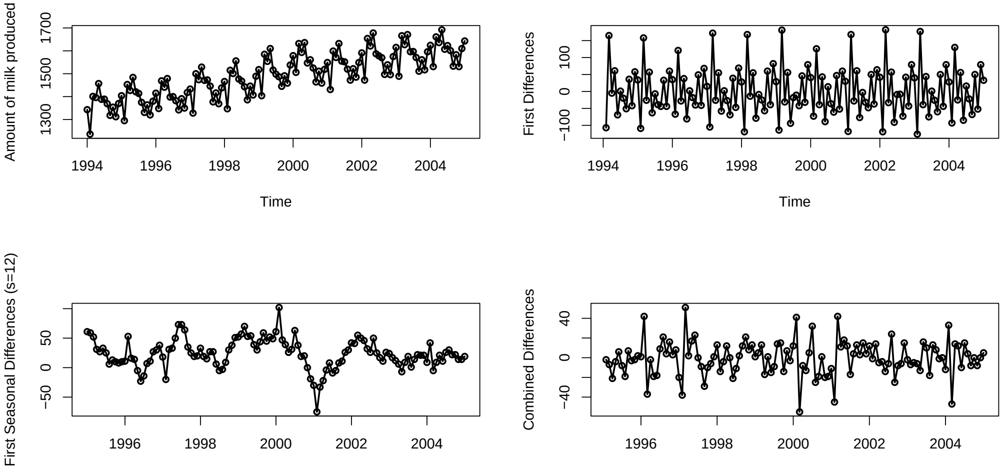

Series milk
Partial ACF
-0.6
-0.2
0.2
0.6
0
1
2
3
4
Lag


Series as.vector((diff(milk)))
Partial ACF
-0.8
-0.4
0.0
0.4
0
10
20
30
40
50
Lag


Series as.vector(diff((milk), 12))
Partial ACF
-0.2
0.2
0.6
0
10
20
30
40
50
Lag


Series as.vector(diff(diff(milk), 12))
Partial ACF
-0.4
-0.2
0.0
0.2
0
10
20
30
40
50
Lag

Step 3: Taking Differences



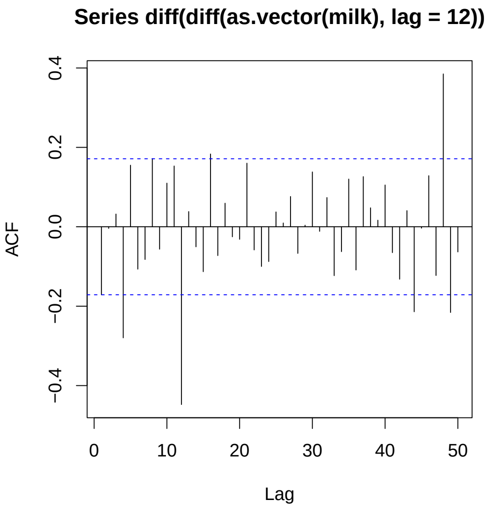
ADF test

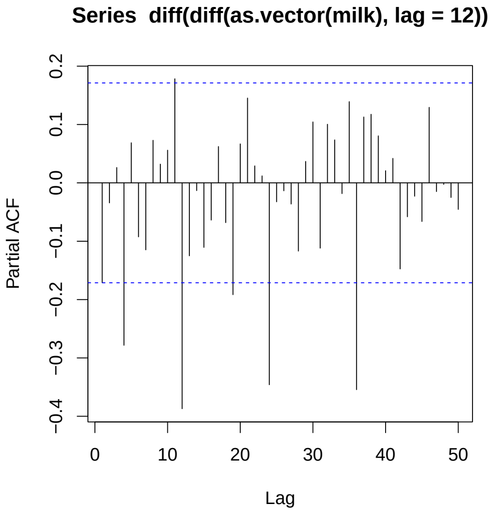
```
> adf.test(diff(diff(milk),lag=12))        #In package tseries

    Augmented Dickey-Fuller Test

data:  diff(diff(milk), lag = 12)
Dickey-Fuller = -5.6125, Lag order = 4, p-value = 0.01
alternative hypothesis: stationary
```

Step 3: Model Specification





Sample ACF and PACF Analysis

- The sample ACF for $\{\nabla\nabla_{12} Y_t\}$ has a pronounced spike at seasonal lag $k = 12$ and one at $k = 48$, but none at $k = 24$ and $k = 36$.
- The sample PACF for $\{\nabla\nabla_{12} Y_t\}$ displays pronounced spikes at seasonal lags $k = 12, 24,$ and 36.

Model Selection Considerations

The last two observations suggest the following choices:

- $(P, Q) = (0, 1)$ if one is willing to ignore the ACF at $k = 48$.

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- $(P, Q) = (3, 0)$, if one strongly considers the sample PACF. There is little happening around seasonal lags in the ACF, and the ACF at $k = 1$ is borderline. We set $p = 0$ and $q = 0$.

Emerging Models

Two strong model possibilities:

- $(P, Q) = (0, 1)$: $MA(1)_{12}$ model for $\{\nabla\nabla_{12} Y_t\}$.
- $(P, Q) = (3, 0)$: $AR(3)_{12}$ model for $\{\nabla\nabla_{12} Y_t\}$.

Model Comparison

Careful examination of both models shows that the $\mathrm{AR}(3)_{12}$ model provides a much better fit than the $\mathrm{MA}(1)_{12}$ model:

- The $\mathrm{AR}(3)_{12}$ model yields a smaller AIC.
- It also has a smaller estimate of the white noise variance.
- Superior residual diagnostics: the Ljung-Box test strongly rejects the $\mathrm{MA}(1)_{12}$ model for $\{\nabla\nabla_{12} Y_t\}$ at all lags.

Final Model Choice

- For illustration, we tentatively adopt an $ARIMA(0,1,0) \times ARIMA(3,1,0)_{12}$ model for the milk production data.

Step 4: Model Fitting

```
> milk.arima010.arima310 = arima(milk, order=c(0,1,0),
                             method='ML', seasonal=list(order=c(3,1,0),
                             period=12))
> milk.arima010.arima310

  Coefficients:
             sar1      sar2      sar3
             -0.9133   -0.8146   -0.6002
   s.e.      0.0696     0.0776   0.0688
sigma^2 estimated as 121.4:
log likelihood = -512.03, aic = 1030.05
```

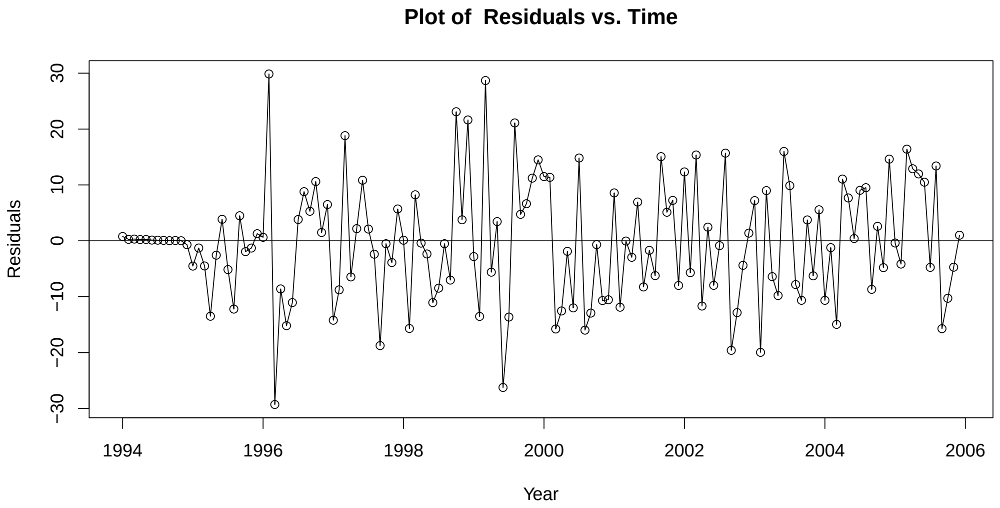
Step 4: Model Fitting

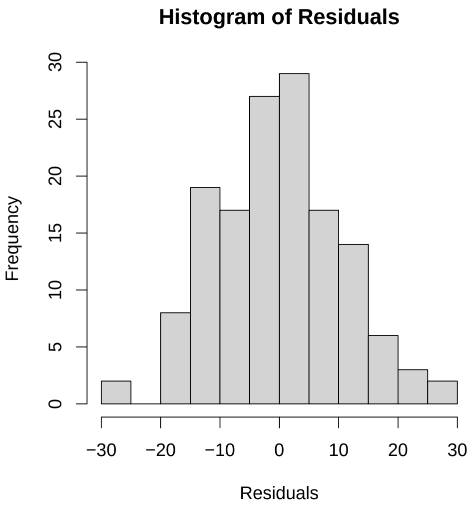
- The fitted model is:

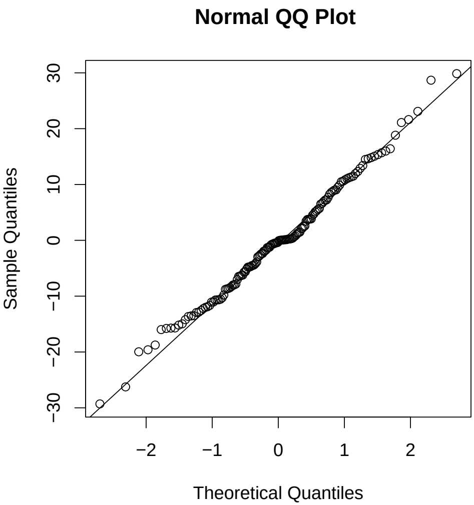
$$(1 + 0.9133B^{12} + 0.8146B^{24} + 0.6002B^{36})(1 - B)(1 - B^{12})\,Y_t = e_t.$$

- The white noise variance estimate is $\hat{\sigma}_e^2 \approx 121.4$.

Step 5: Model Diagnostics



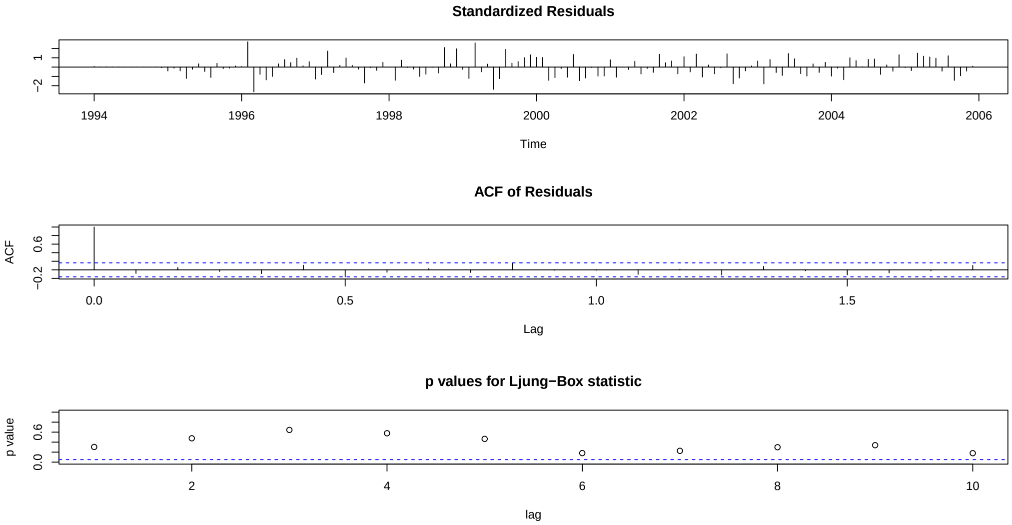

Histogram of Residuals
Frequency
Residuals
0
5
10
15
20
25
30
−30
−20
−10
0
10
20
30


Normal QQ Plot
Sample Quantiles
Theoretical Quantiles
−30
−20
−10
0
10
20
30
−2
−1
0
1
2

```
> shapiro.test(Res)

	Shapiro-Wilk normality test

data:  Res
W = 0.9926, p-value = 0.6618
```

```
> runs(Res)
$pvalue
[1] 0.112

$observed.runs
[1] 63

$expected.runs
[1] 72.98611

$n1
[1] 73

$n2
[1] 71

$k
[1] 0
```

```
> tsdiag(milk.arima010.arima310)
```



Step 6: Forecasting

```
> predict(milk.arima010.arima310,n.ahead = 6)
$pred
          Jan      Feb      Mar      Apr      May      Jun
2006 1702.409 1584.302 1760.356 1728.246 1783.487 1698.330

$se
          Jan      Feb      Mar      Apr      May      Jun
2006 11.01761 15.58125 19.08305 22.03521 24.63612 26.98751
```

```
>  plot(milk.arima010.arima310,n1=c(2003,1),n.ahead=24,
       col='red',type='b',pch=16,ylab="Milk Prodction")
```